

A Jogged Dislocation Governed Strengthening Mechanism in Nanotwinned Metals

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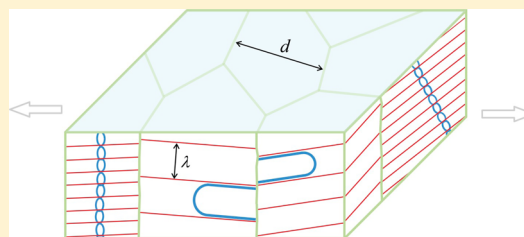
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S Supporting Information

ABSTRACT: Atomistic simulations reveal a new and unique strengthening mechanism in nanotwinned metals governed by the collective motion of multiple necklace-like extended jogged dislocations. This mechanism prevails in a columnar-grained nanotwinned metal subject to an external stress parallel to the twin planes, provided the twin boundary spacing falls below a critical value. A theoretical model based on the depinning of unit jogs on twin planes is proposed to determine the flow stress associated with this deformation mechanism and is shown to be in agreement with atomistic simulations.

KEYWORDS: Jogged dislocation, threading dislocation, twin boundary, plastic deformation, atomistic simulation



Nanotwinned metals with a high density of nanoscale twins have attracted worldwide attention due to their unique mechanical and physical properties including ultrahigh strength and hardness, good tensile ductility and strain hardening, superior fatigue and wear resistance, and excellent electrical conductivity.^{1–12} A number of deformation mechanisms and their associated effects on the strength of nanotwinned metals have been identified.^{3,5–14} It has been shown that twin boundaries (TBs) in the nanotwinned metals provide adequate barriers to inclined dislocation slip, leading to the classical Hall–Petch behavior of material strengthening with reduced TB spacing,^{6,11,12} similar to the traditional grain boundary (GB) strengthening in polycrystalline metals.^{15–17} Recently, it has also been reported that a gradient hierarchical nanotwinned microstructure can be introduced into the twinning-induced plasticity steel through pretorsion, leading to a dramatic enhancement of strength at no ductility trade-off.¹⁸ This unexpected evasion of strength-ductility trade-off in steel has been attributed to interactions between dislocations and TBs and among hierarchical twin structures.¹⁸ Experimental and computational studies^{6,8,9} on deformation in nanotwinned metals have revealed a transition in deformation mechanism from the Hall–Petch strengthening to a dislocation-nucleation-governed softening mechanism that involves the nucleation and motion of twinning partial dislocations along the twin planes below a critical twin thickness. In this softening mechanism, a continuous slip of twinning partial dislocations on twin planes leads to thinning and even annihilation of twin lamellae, a process often referred to as detwinning.^{8,9,19}

The above two typical deformation mechanisms (dislocation slip inclined/parallel to TBs) have been experimentally

observed in nanotwinned polycrystalline metals with equiaxed grains^{4,8,9} where TBs and GBs are randomly oriented, as well as nanotwinned nanopillars/nanowires^{20,21} with only TBs but no other general GBs. More recently, nanotwinned metals with columnar grains have been fabricated through direct current electrodeposition.^{11,13,14} Such samples have a strong {111} out-of-plane texture and preferentially oriented nanoscale twins with TBs nearly perpendicular to the axial directions of the columnar grains.^{13,14} Columnar-grained nanotwinned Cu exhibits a remarkable plastic anisotropy¹⁴ and excellent fatigue properties,²² as their deformation behaviors are strongly dependent on the TB spacing and orientation. These phenomena provide a promising foundation to optimize mechanical properties of nanotwinned materials by tailoring their twin microstructures. In the columnar-grained nanotwinned metals, when the loading direction is oriented in parallel to the twin planes, both dislocation pile-ups due to TB impediment and easy glide of partial dislocations along the twin planes are to a large extent suppressed, leading to activation and glide of threading dislocations confined between neighboring TBs.¹³ A similar mechanism has been previously reported in nanolayered metallic composites where the plastic deformation is strongly associated with the glide of threading dislocation loops confined in the nanolayers.^{23,24} In these lamellar materials, the strength is enhanced as the layer thickness is reduced, which can be described by the so-called confined layer

Received: May 10, 2014

Revised: August 7, 2014

Published: August 18, 2014

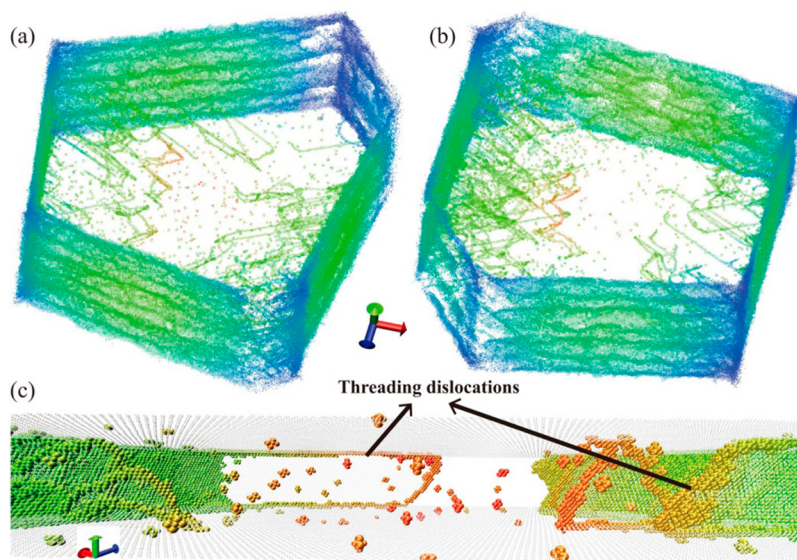


Figure 1. (a, b) Dislocation structures in two representative grains of the simulated sample with $d = 50$ nm and $\lambda = 5.01$ nm at 6.18% strain. (c) Atomic structures of threading dislocations in one twin lamella via position-based coloring. It is observed that two misfit segments lie on the twin planes (containing atoms in silver), and the threading segment is confined in the twin lamellae.

slip (CLS) model of threading dislocations.²³ In the present work, by using large-scale atomistic simulations, we report yet another mechanism of plastic deformation which involves the collective motion of multiple extended jogged dislocations that span across and are pinned at multiple twin planes. It will be shown that this mechanism governs the strengthening of nanotwinned metals as the twin thickness is reduced to around 1 nm.

Fully three-dimensional polycrystals with three different mean grain sizes, $d = 20, 30$, and 50 nm, are used in the molecular dynamics (MD) simulations. Each sample contains four columnar grains, where the coherent TBs are perpendicular to the axial direction of grains (i.e., the thickness direction). Hence, the simulation samples are (111)-textured with tensile loading applied along the twin-planes, similar to the experiments.¹³ The in-plane misorientations between neighboring grains are larger than 30° to generate high-angle GBs. For samples with $d = 20$ and 30 nm, the thickness is about 25 nm, and TB spacing λ varies from 0.63 to 12.50 nm. For the $d = 50$ nm samples, the thickness is about 30 nm and TB spacing is taken to be 0.83 and 5.01 nm. The largest simulated sample contains 25.6 million atoms. During simulations, periodic boundary conditions are imposed in all three directions. Uniaxial tension is applied parallel to the twin planes. The embedded atom method potential for Cu is employed to describe the interatomic interactions,²⁵ and a typical MD time step of 1.0 fs is used. Before stretching, the simulation system is equilibrated for 200 ps at room temperature and zero pressure via the NPT ensemble.^{26,27} After equilibration, the sample is uniaxially stretched at a constant strain rate of 2×10^8 s⁻¹. During tension, the temperature is kept at 300 K, and the pressures in the two nontensile directions are controlled at zero through NPT ensemble. The common neighbor analysis (CNA) method is adopted to identify defects that arise in the simulated sample.^{28,29} Colors are assigned to atoms according to their local crystal order: gray stands for perfect atoms, red for atoms in stacking faults or TBs, green for atoms in dislocations or GBs, blue for atoms near vacancies, and yellow for fully disordered atoms. Accordingly, single red layers

represent TBs, while double red layers indicate intrinsic stacking faults. Combining the CNA coloring scheme, we also adopt a position-based coloring to generate perspective 3D effect,¹⁴ where the colors represent the distance of atoms from the center of a simulated grain.

Figure 1a and b show dislocation structures in two representative grains in the sample with $d = 50$ nm and $\lambda = 5.01$ nm at 6.18% strain. Most of the dislocation lines in Figure 1a and b exhibit a hairpin-like shape. This kind of dislocation loops is generally referred to as threading or channeling dislocations. Figure 1c shows the atomic configuration of typical dislocation loops confined in twin lamellae by position-based coloring. As illustrated in Figure 1c, a threading dislocation has two misfit segments on the neighboring twin planes and one threading segment spanning across the twin lamella. During the simulation, threading dislocation loops are found to initiate from GBs, glide through grain interior, and become finally absorbed at the opposite GBs. Such deformation mechanism prevails in the samples with $d = 20$ and 30 nm and TB spacing larger than 3.75 nm. In thin films^{30,31} or nanolayered metallic composite,^{23,24} the nucleation and motion of threading dislocation is known to be a controlling mechanism responsible for plastic deformation, resulting in enhanced strength with decreasing lamellar thickness. In the CLS model suitable for describing such size strengthening effect in thin films, the yield strength is expressed as^{30,31}

$$\sigma_{\text{cls}}^e = M \left[\frac{\mu b \sin \varphi}{8\pi\lambda} \left(\frac{4 - \nu}{1 - \nu} \right) \ln \left(\frac{\alpha\lambda}{b \sin \varphi} \right) + \tau_0 \right] \quad (1)$$

where λ is the film thickness, μ the shear modulus, ν the Poisson's ratio, τ_0 the resistance to dislocation glide from lattice and other obstacles, M the Taylor factor, φ the angle between slip plane and the layer interface, b the Burgers vector of dislocation, and α a coefficient representing the extent of dislocation core. For a nanolayered metallic composite, the CLS model is modified as²³

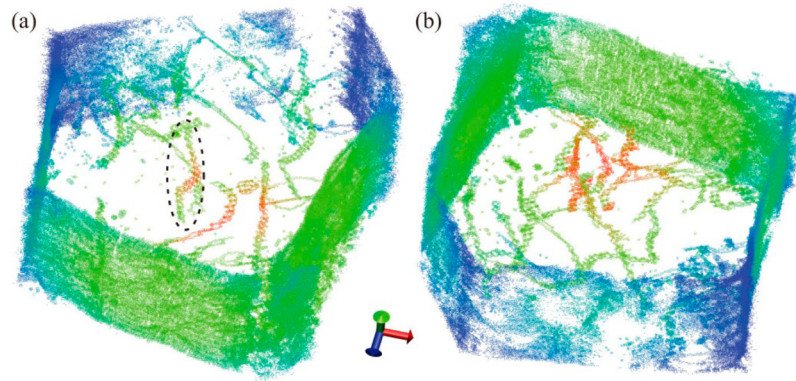


Figure 2. (a, b) Dislocation structures in two representative grains of the simulated sample with $d = 50$ nm and $\lambda = 0.83$ nm at 10.30% strain. Multiple extended dislocations connect with each other, constituting a jogged dislocation.

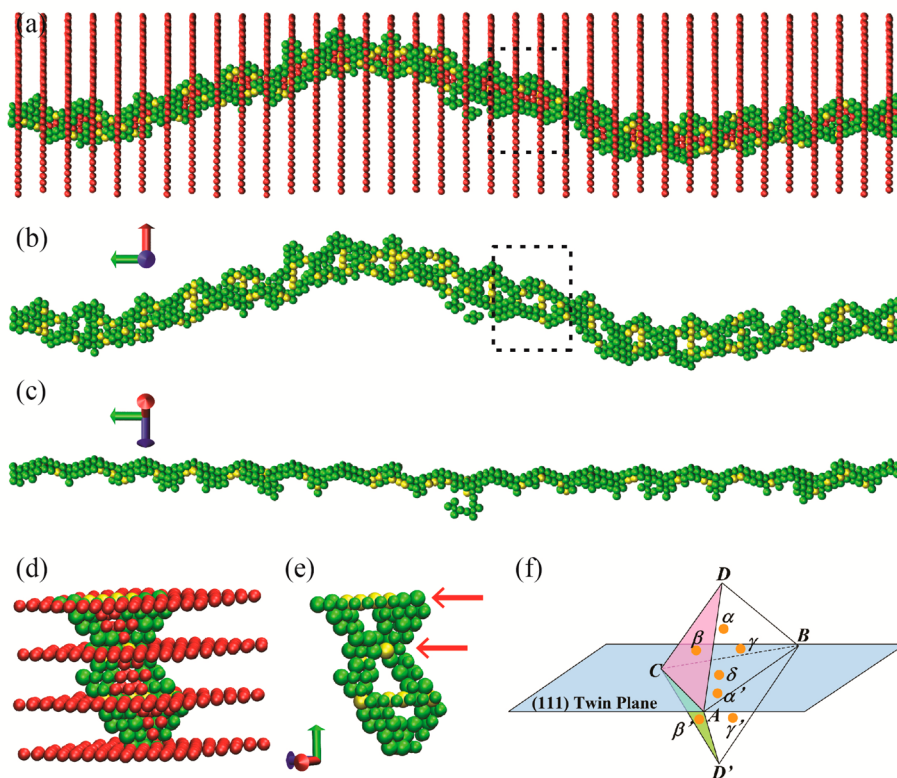


Figure 3. (a–c) Detailed atomic structures of a jogged dislocation (circled in Figure 2a) in CNA coloring from different views. The jogged dislocation exhibits a zigzag shape from a certain perspective. (d, e) Magnification of the segments within the dashed line frames in a and b. The extended and constricted jogs of one segment are indicated by the red arrows. (f) Double Thompson tetrahedron representation in the twin–matrix system. The upper tetrahedron represents the slip systems for the matrix, while the lower one for the twin.

$$\sigma_{\text{cls}}^{\text{r}} = M \frac{\mu b \sin \varphi}{8\pi\lambda} \left(\frac{4 - \nu}{1 - \nu} \right) \ln \left(\frac{\alpha\lambda}{b \sin \varphi} \right) - \frac{\gamma}{\lambda} + \frac{\mu b}{l(1 - \nu)} \quad (2)$$

after the influence of interfaces is accounted for. The first term on the right-hand side of eq 2, similar to that of eq 1, represents the stress required to drag the misfit segments of a threading dislocation. The second term represents the prestress associated with the interfaces, with γ being the interfacial energy (J/m^2). The third term reflects the interaction between a glide threading dislocation and a pre-existing array of misfit dislocations on the interfaces, with l denoting the mean dislocation spacing. For a twin lamellae with thickness on the

order of several nanometers, the interfacial energy of TB being $\gamma_{\text{T}} = 24.0$ mJ/m^2 for Cu,²⁵ the contribution from the second term in eq 2 to strength is negligible. Therefore, regarding the last two terms in eq 2 as a constant σ_{d} independent of TB spacing, one may rewrite eq 2 as

$$\sigma_{\text{cls}}^{\text{r}} = M \frac{\mu b \sin \varphi}{8\pi\lambda} \left(\frac{4 - \nu}{1 - \nu} \right) \ln \left(\frac{\alpha\lambda}{b \sin \varphi} \right) + \sigma_{\text{d}} \quad (3)$$

In subsequent analysis, we will use eq 3 to account for the strengthening due to threading dislocations confined in twin lamellae.

Figure 2 shows deformation patterns in two representative grains in the sample with $d = 50$ nm and $\lambda = 0.83$ nm at 10.30%

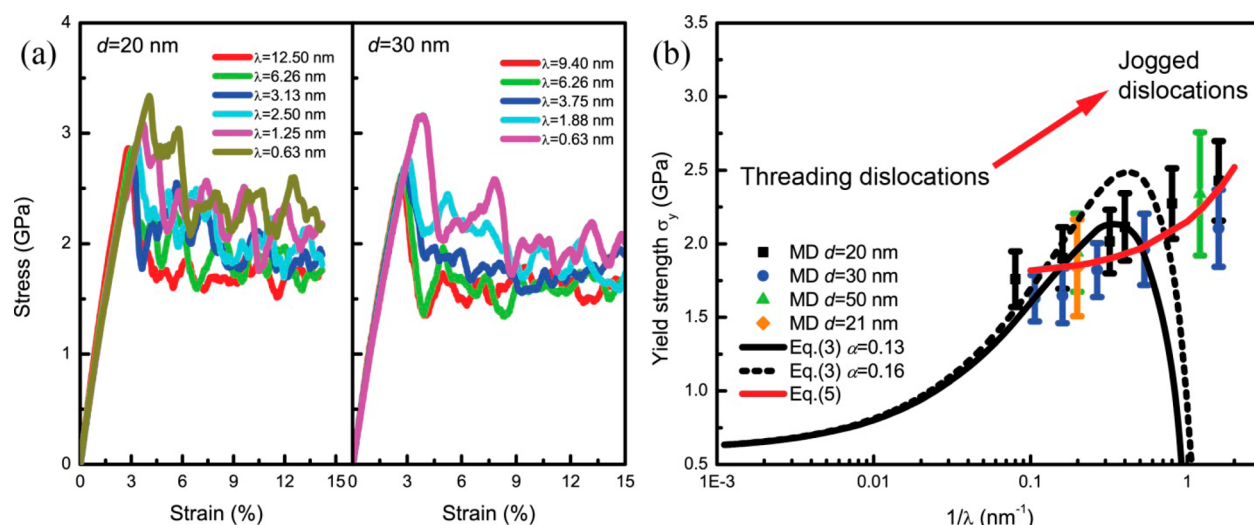


Figure 4. (a) Stress–strain curves of simulated samples with $d = 20$ and 30 nm and different TB spacings. (b) The twin size dependence of material strength. There exists a transition from threading dislocation to jogged dislocation governed deformation regimes with reduced TB spacing. The orange diamond is from ref 35, and all other data are from the current study.

strain, which is very distinct from those shown in Figure 1. It is observed that the plastic deformation is governed by activities of necklace-like dislocations extended over multiple twin planes, which we refer to as jogged dislocations. Each of these jogged dislocations is nucleated as a whole from a GB (see Supplementary Movie), rather than through assembly or interactions of separately emitted dislocations. The nucleation process should be assisted by diffusion and coalescence of vacancies.³² (see Supplementary Movie). Figure 3a and b show the detailed atomic structures of a single jogged dislocation spanning across multiple twin planes, while Figure 3c and d show the magnification of three consecutive segments of the jogged dislocation. The dislocation pattern consists of multiple extended dislocations, each bounded by two twin planes. Every extended dislocation contains a pair of leading and trailing partial dislocations separated by a stacking fault. Because of the geometrical constraints, each extended dislocation is terminated at a twin plane and joined with another dislocation in the neighboring twin through a common segment in the twin plane. Based on the double Thompson tetrahedron representation in Figure 3e, the leading and trailing partials in matrix have Burgers vectors of $A\beta$ and βC , while those in twin have Burgers vectors of $A\beta'$ and $\beta'C$. Thus, the common segment in the twin plane has the Burgers vector of a stair-rod dislocation $\beta'\beta$. Such a stair-rod segment possesses the characteristic of a unit jog and is pinned at the twin plane. Depending on its Burgers vector and the corresponding line tension, the unit jog may be either of extended or constricted type, as illustrated in Figure 3d. Similar dislocation structures have been reported in single crystalline Ni, where double unit jogs of one interplanar spacing were artificially introduced in atomistic modeling.³³ Some zigzag slip bands across a few TBs have been experimentally observed in fatigue of columnar-grained nanotwinned Cu,²² which might be attributed to the jogged dislocation mechanism during cyclic deformation. In the simulated samples with grain sizes $d = 20$ and 30 nm, the same deformation mechanism governed by jogged dislocations is observed at TB spacings below 3.75 nm. Interestingly, the motion of jogged dislocation involves the collective movements of multiple extended dislocations which are linked by numerous unit jogs pinned at twin planes. As the extended part of a jogged dislocation is

driven forward by the applied stress, the unit jogs undergo depinning at the twin planes, which is assisted by diffusion of point defects.³⁴ Before proposing a strengthening model based on the jogged dislocation mechanism, we first explore the transition in the governing deformation mechanisms at decreased twin size.

Comparison between Figures 1 and 2 demonstrates that there exists a distinct transition from threading dislocation to jogged dislocation governed deformation mechanisms as the twin thickness is reduced. It would seem that the sharing of misfit segments by forming the extended jogged dislocations should be energetically favorable regardless of the twin spacing. However, a large stress is still required to drag these jogged dislocations along multiple twin planes. In comparison, the critical stress required to move a threading dislocation with two misfit segments is inversely proportional to the TB spacing, as indicated in eq 3. At a relatively large TB spacing, only moderate stress is needed to activate threading dislocations in individual twin/matrix lamellae, and this dominates the plastic deformation. When the TB spacing falls below a critical size, the critical stress for threading dislocation operation has reached the value required to move extended jogged dislocations along multiple twin planes, and the two misfit segments of a threading dislocation are so close to each other that the entire threading dislocation structure becomes energetically unfavorable. As a result, the extended jogged dislocations are activated and move collectively on multiple twin planes. Figure 4a shows typical stress–strain curves of the samples with $d = 20$ and 30 nm. If we take the average of stresses after yielding as the flow stress, it is seen from Figure 4b that the flow stress gradually increases with decreasing TB spacing. This means that the presence of TBs always strengthens the columnar-grained nanotwinned metals if the external loading is parallel to the TBs, in contrast to the softening phenomenon previously revealed in the equiaxed nanotwinned metals in the similar range of twin thickness.^{6,8,9} We find that the CLS model in eq 3 fails to correlate the size dependence of materials strength in Figure 4b. In fact, the prediction from eq 3 is found to decrease when TB spacing λ falls below a certain value, as shown in Figure 4b. As the layer thickness becomes comparable to the cutoff radius of

dislocation core, the first term in eqs 1 and 3 becomes negative due to the logarithmic component, causing a stress drop which is contradictory to our simulation results. Relevant parameters used in eq 3 are summarized in Table 1.

Table 1. Parameters Used in Equations 3 and 5 for Cu

μ (GPa)	ν	b (nm)	φ	α^a
48	0.35	0.147	70.5°	0.13, 0.16
M	σ_d (GPa)	τ_0 (GPa)	b_f (nm)	f_{pin} (pN)
3	0.60	0.59	0.255	31.48

^a α is a parameter related to dislocation core extension. Two tentative values of $\alpha = 0.13$ and 0.16 are listed. The former is from a fitting to MD data, while the latter comes from an experimental study on the tensile behavior of columnar-grained nanotwinned Cu.¹³

The deformation patterns shown in Figure 2 imply that the strengthening of a columnar-grained nanotwinned metal, as the TB spacing is reduced to around 1 nm (comparable to the cutoff radius of dislocation core), must be associated with the collective motion of jogged dislocations. Consequently, the CLS model is no longer valid at sufficiently small TB spacings (see Figure 4b) due to a transition in deformation mechanism. To explain the above TB strengthening behavior, especially in the small twin-thickness range, we propose a simple model based on the motion of jogged dislocation and depinning of unit jogs on twin planes. For every constituent loop in the jogged dislocation, the pinning force f_{pin} exerted by TBs on the unit jog during its motion is compensated by the associated Peach–Koehler force acting on the dislocation. Accordingly, the equilibrium equation of a dislocation should be given by,

$$f_{pin} = (\tau - \tau_0)b_f\lambda \quad (4)$$

where τ is the resolved shear stress, τ_0 the friction stress, and b_f the magnitude of Burgers vector of dislocation. Here the subscript “f” in b is used to emphasize the full-dislocation nature of the jogged dislocation. Solving eq 4 and considering the Taylor factor M , we obtain the following steady-state flow stress σ_{jog} due to the depinning process,

$$\sigma_{jog} = M \left(\frac{f_{pin}}{b_f\lambda} + \tau_0 \right) \quad (5)$$

As shown in Figure 4b, as the TB spacing falls below 3.75 nm, the strength enhancement can be well-described by eq 5 with $f_{pin} = 31.48$ pN and $\tau_0 = 0.59$ GPa for the case of Cu. On the basis of our simulations and the theoretical predictions in Figure 4b, the transition from threading to jogged dislocations occurs around 2–5 nm. Within this regime, the threading and jogged dislocations could coexist during plastic deformation. As the TB spacing is reduced across this size range, the dominant mechanism of plastic deformation gradually changes from a threading dislocation dominated regime to one that is fully dominated by jogged dislocations throughout the grain interiors.

While the jogged dislocations reported here have been discovered in the columnar-grained nanotwinned materials below a critical TB spacing, an interesting question is whether this unique deformation mechanism could exist under more general conditions. The jogged dislocations are expected to dominate plastic deformation even if the loading direction is slightly tilted from the twin planes (see Supplementary Figure

1), as long as the resolved shear stress on the twin planes is not sufficient to activate twinning partial dislocations. In many nanolamellar/nanolaminated/nanolayered materials including thin films, coatings, and nanolayered and nanotwinned systems subject to loading parallel to the layers, threading dislocations have been widely reported and investigated.^{13,14,23,24,31} From an energetic point of view, if the characteristic layer thickness is reduced below a critical length, the threading dislocations could become unfavorable due to the close interaction of their misfit segments lying on adjacent interfaces. In this case, the jogged dislocations are more stable due to their constricted and compact structures. This analysis would suggest that the jogged dislocation mechanism could be generally prevalent in nanolayered materials, provided that the characteristic layer thickness is sufficiently small and the applied stress is nearly parallel to the interfaces.

Recent fatigue experiments²² on columnar-grained nanotwinned Cu have revealed a mysterious type of distinct zigzag slip bands spanning across multiple TBs, exhibiting a well-defined 45° angle relative to the loading axis, which might be a manifestation of jogged dislocation mechanism under cyclic deformation at macroscopic scale. However, due to the apparent differences in length scale, stress level, and loading mode, further study about influence of cyclic loading on jogged dislocation mechanism is necessary to better correlate the present atomistic deformation mechanism with the macroscopic deformation in the experiments. Also, nanotwinned metals containing angstrom-scaled twins have been successfully synthesized in the laboratory.^{21,36} These developments are calling for further investigations of the jogged dislocation mechanism.

In summary, using a large-scale atomistic simulation, we have revealed a new and unique jogged dislocation governed deformation mechanism in nanotwinned metals. This mechanism has been demonstrated in a columnar-grained nanotwinned metal subject to external loading parallel to the twin planes for TB spacing below a critical value. In this system, for TB spacing larger than a critical value, plastic deformation is governed by the nucleation and propagation of threading dislocations, which can be described by the CLS model. When the TB spacing falls below around 1 nm, the plastic deformation is dictated by the activities of jogged dislocations, leading to continuous strengthening with reduced TB spacing. Such a dislocation mechanism involves the collective behavior of multiple extended jogged dislocations containing unit jogs bounded by neighboring twin planes. Based on the critical insight of depinning of unit jogs in atomistic simulations, we have proposed a theoretical model to describe the strengthening law of jogged dislocations. The predictions from the theoretical model are shown to agree well with the simulation results. The new mechanism not only underscores a unique interaction mode between TBs and dislocations but also enriches our understanding of plastic deformation mechanisms in nanotwinned materials.

■ ASSOCIATED CONTENT

📺 Supporting Information

Movie showing the nucleation and propagation of jogged dislocations in a representative grain of the simulated samples with a grain size of 50 nm and a TB spacing of 0.83 nm, as well as a figure confirming the mechanism of jogged dislocations for TBs slightly tilted away from the loading direction. This

material is available free of charge via the Internet at <http://pubs.acs.org>.

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H.Z. and X.L. contributed equally to this work.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

X.L. and H.G. acknowledge financial support by NSF through Grant CMMI-1161749 and MRSEC Program through award DMR-0520651 at Brown University and computational support from NICS through Grant MS090046. H.Z., S.Q., and W.Y. acknowledge the support from Zhejiang University for H.Z.'s thesis work and also financial support from NSFC through Grant Nos. 11172264, 11222218, and 11321202. X.L. also acknowledges support from the Thousand Young Talent Program of China and NSFC through Grant No. 11372152.

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